

An Overview of the Proposed PE Study

Main architecture, staged training, and plan for diagnosis and prognosis

– – Quoc Duy – –

This is a overview picture about project PE and step by step experiment to analysis

I will conduct the study sequentially, starting with **diagnosis**, followed by **prognosis**, and finally **counterfactual analysis**.

0. Study scope: two PE tasks

- Dataset:
 - + STANFORD INSPECT data (**train/validation/internal test**)
 - + Turkey (**external test only**)
 - + RSPECT (from kaggle with multi-label)
- Mindset design architecture
 - + *Organ-specific deep feature extraction and fusion instead of whole-volume learning. Because I think model also focus on importance organ.*
- Two Complementary Tasks for Pulmonary Embolism (Final Expect Model):

Task A: PE Diagnosis

Objective: Detect pulmonary embolism and characterize its imaging phenotype.

Inputs:

- + Full 3D CTPA volume
- + Anatomy-aware features from the heart, pulmonary arteries, and lungs

Expected outputs (label using):

- + pe_positive: **positive / negative (priority)** - main task
- + pe_acute_nlp
- + pe_subsegmentalonly_nlp

Task B: Multimodal PE Prognosis

Objective: Estimate the clinical risk and short-term outcome of patients with acute PE.

Inputs:

- + Full 3D CTPA and anatomy-aware image features
 - + Clinical and EHR variables
 - + Missingness indicators
 - + ~~PESI or sPESI score~~
- dataset no contain enough variables to calculate the PESI score.**

Expected outputs:

- **Primary: 30-day mortality in PE patients.**
- Secondary: 30-day mortality in the all-comer CTPA cohort.
- Exploratory: 6-/12-month mortality, readmission and 12-month PH.

1.Run init basic baseline to find mini dataset in INSPECT:

For the small-scale setting, we use a subset of the data with three diagnostic labels and one primary prognosis endpoint for PE-positive patients.

Using resnet152 and swinUNETR with 3 head to evaluate performance in diagnosis task and prognosis (1 task) 30-day mortality in PE patients with full data



Based on error analysis and model performance, select 500 representative samples from the dataset to create a **standardized test set for evaluating the proposed model instead of using full INSPECT.**

All next step using mini data to evaluate to find best method.

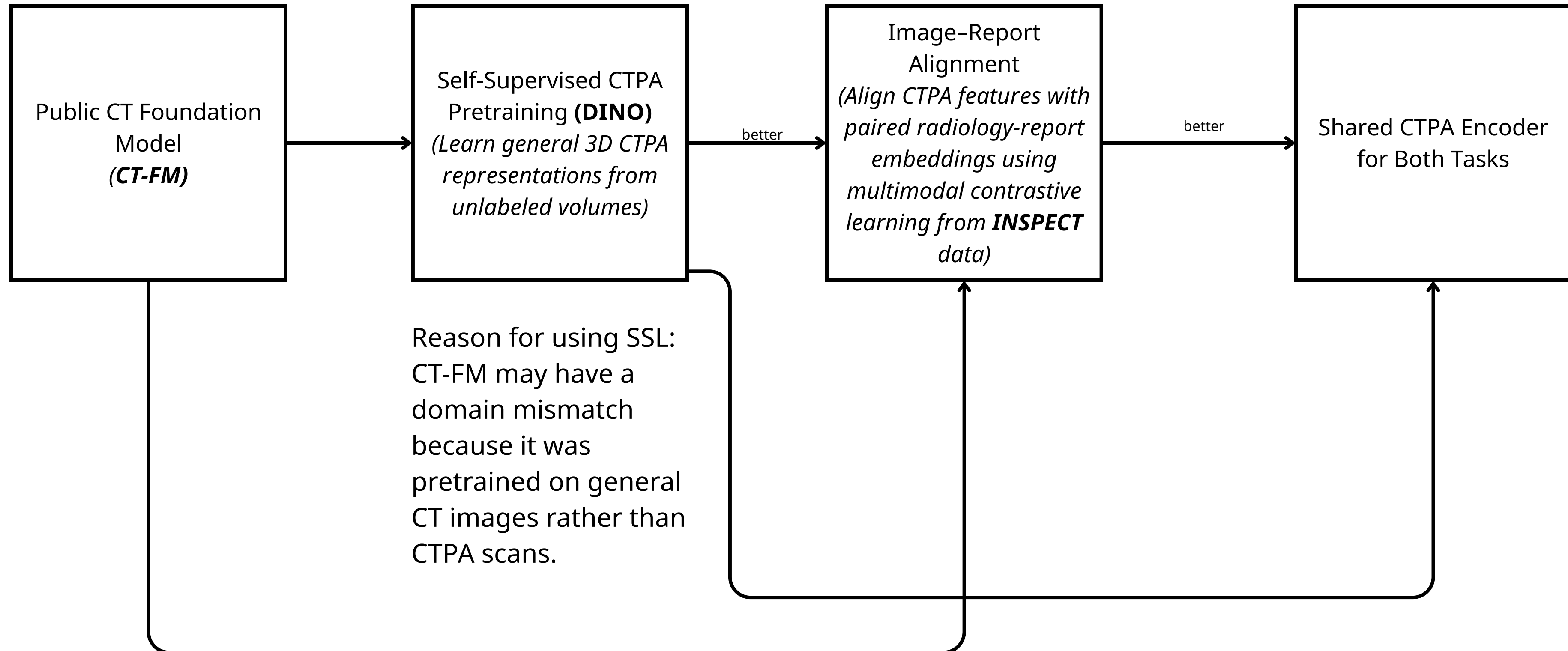


- Evaluate CT-FM in a zero-shot setting to eval the domain gap between CT and CTPA.
- Evaluate PENet in a zero-shot setting.
- Then, fine-tune and evaluate both CT-FM and PENet.

Output stage 1: Task		diagnosis (3 tasks)	Prognosis (1 task)
model \ Label		PE+- , pe_acute pe_subsegmenta lonly	30-day mortality in PE patients.
resnet152 (run full data)			
swinUNETR (run full data)			
CT-FM (mini data with fintune and test set)	zero-shot if having weight task		
	Finetune		
PENET (mini data with fintune and test set)	zero-shot		
	Finetune		

2. Shared foundation before task-specific models with mini dataset (train/val/test)

PE Task Model: Conduct a stepwise ablation study and remove components that do not improve or that degrade performance. Compare by using weight of model to fine-tune for task diagnosis and prognosis.



2. Shared foundation before task-specific models with mini dataset (train/val/test)

Output Stage 2 with shared encoder:

	diagnosis (3 tasks)	Prognosis (1 task)
model \ Label	PE+- , pe_acute pe_subsegmentalonly	30-day mortality in PE patients.
CT-FM + SSL with basic head		
CT-FM + SSL + Align OR: CT-FM + Align with basic head		

3. Silver label

To support knowledge for model in PE from pretraining silver label with multi-task which can extract from report.

In task Prognosis using pretrain from encoder shared or weight of task diagnosis

To generate silver labels for pretraining, we use MedGemma, which leverages its medical knowledge to extract relevant information from radiology reports.

Stage 3A: DIAGNOSIS TRAINING

Silver-Label Multitask Fine-Tuning with Full architecture model (C_0 if better)

(Extract PE presence, acuity, location, right-heart strain, and associated findings from reports and use them as weak supervision for multitask diagnosis.)

C0 + organ adapters + diagnosis heads => train with silver/gold diagnosis labels => C_diag

Stage 3B: PROGNOSIS TRAINING

C0 or C_diag + prognosis LoRA/adapters + EHR branch => train with outcome labels => C_prog

3. Silver label

Các output kì vọng của stage 3:

+ Compare model with previous model:

	diagnosis (3 tasks)	Prognosis (1 task)
model \ Label	PE+- , pe_acute pe_subsegmentalonly	30-day mortality in PE patients.
shared encoder + sliver label with basic head		

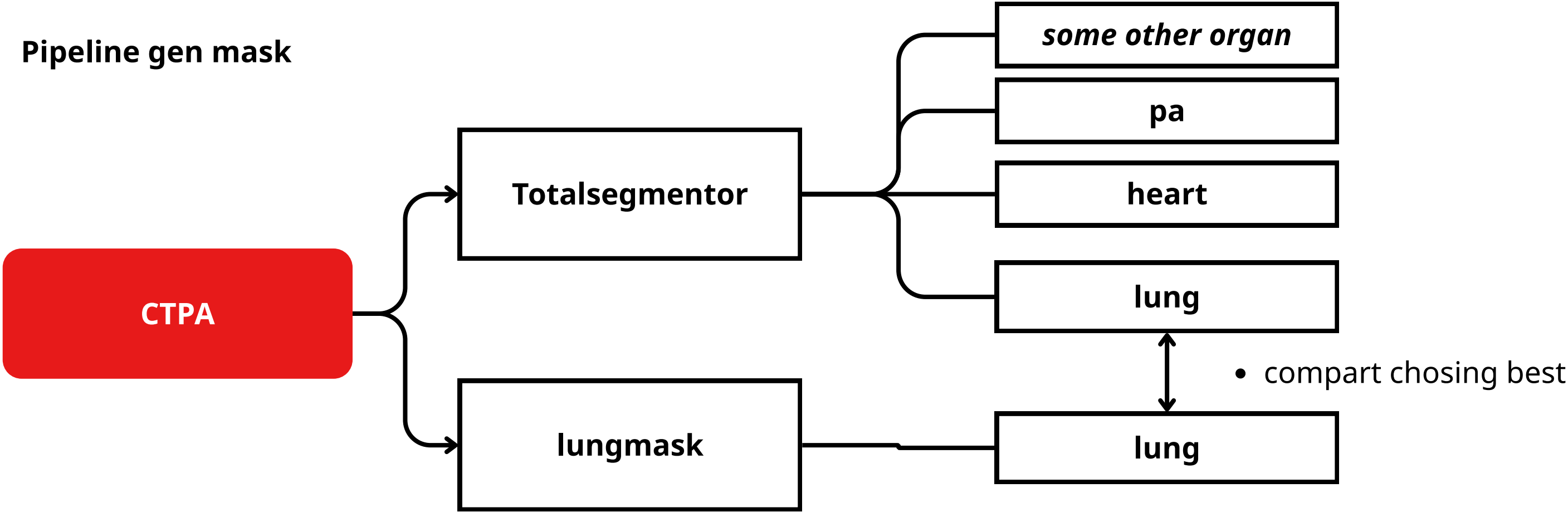
+ And Dataset with sliver lable for INSPECT

4. Design Proposal Architecture

Reason design my architecture:

- The dataset is large, so I propose exploring a Mixture-of-Experts (MoE) architecture in which each expert specializes in a specific organ. Based on my previous experience, MoE models often performed poorly on small datasets but have greater when trained on larger datasets. Therefore, I would like to evaluate this approach.
- Based on the literature review, email discussions, and the professor’s comments regarding organ-specific modeling, I hypothesize that assigning a dedicated expert to each relevant organ could improve organ-specific feature extraction and attention.
- However, the organ masks are automatically generated rather than manually annotated. I use **TotalSegmentator** and **LungMask** for cross-checking, but their available segmentation classes limit the current regions to the **lungs, pulmonary arteries, and heart**.

Pipeline gen mask

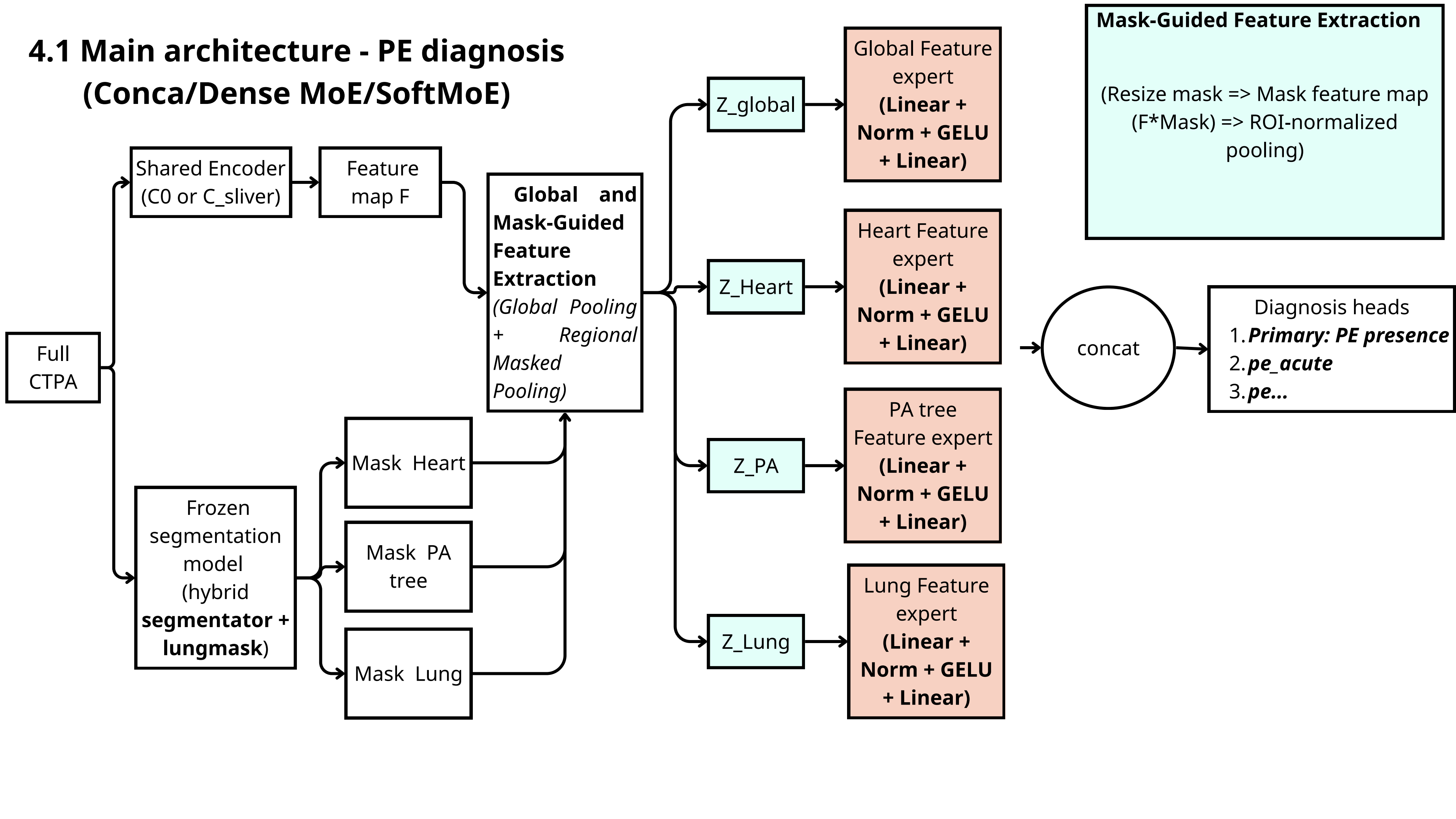


4. Design Proposal Architecture

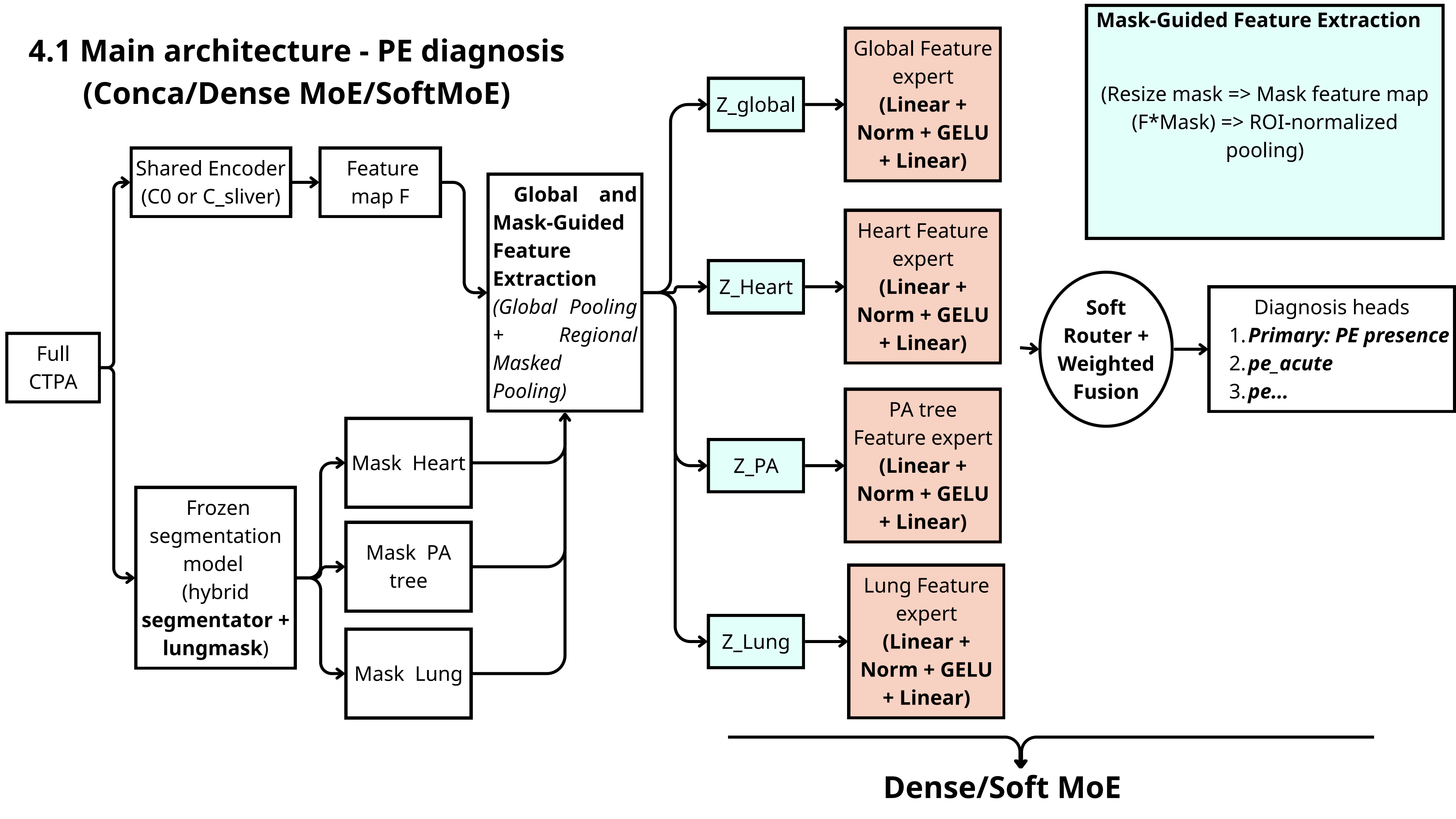
Now, I use pipeline above to gen some masks:

- lung
 - heart
 - strict_heart
 - myocardium
 - **mediastinum**
 - central_pa
 - lung_arteries
 - lung_veins
 - **lung_vessels**
 - airways
 - **pa_tree**
 - hilar_vessels
 - body
 - body_wall
 - rv
 - lv
 - ra
 - la
 - lung_lungmask
- I use an NVIDIA L4 GPU in ggcloud to generate organ masks, but limited memory. I will segment, inspect, and **extract central slices to evaluate mask quality**, as TotalSegmentator was pretrained on general CT data rather than CTPA. The proposed model focuses on three major anatomical regions: **the lungs, heart, and pulmonary artery tree**.
 - If the masks are sufficiently accurate, I will use them for counterfactual masking experiments to quantify the contribution of each organ to PE prediction.
 - Organ-specific ROIs will be created by combining relevant masks, and the voxel values within each ROI will be replaced with random values to measure the resulting change in model performance.

4.1 Main architecture - PE diagnosis (Conca/Dense MoE/SoftMoE)

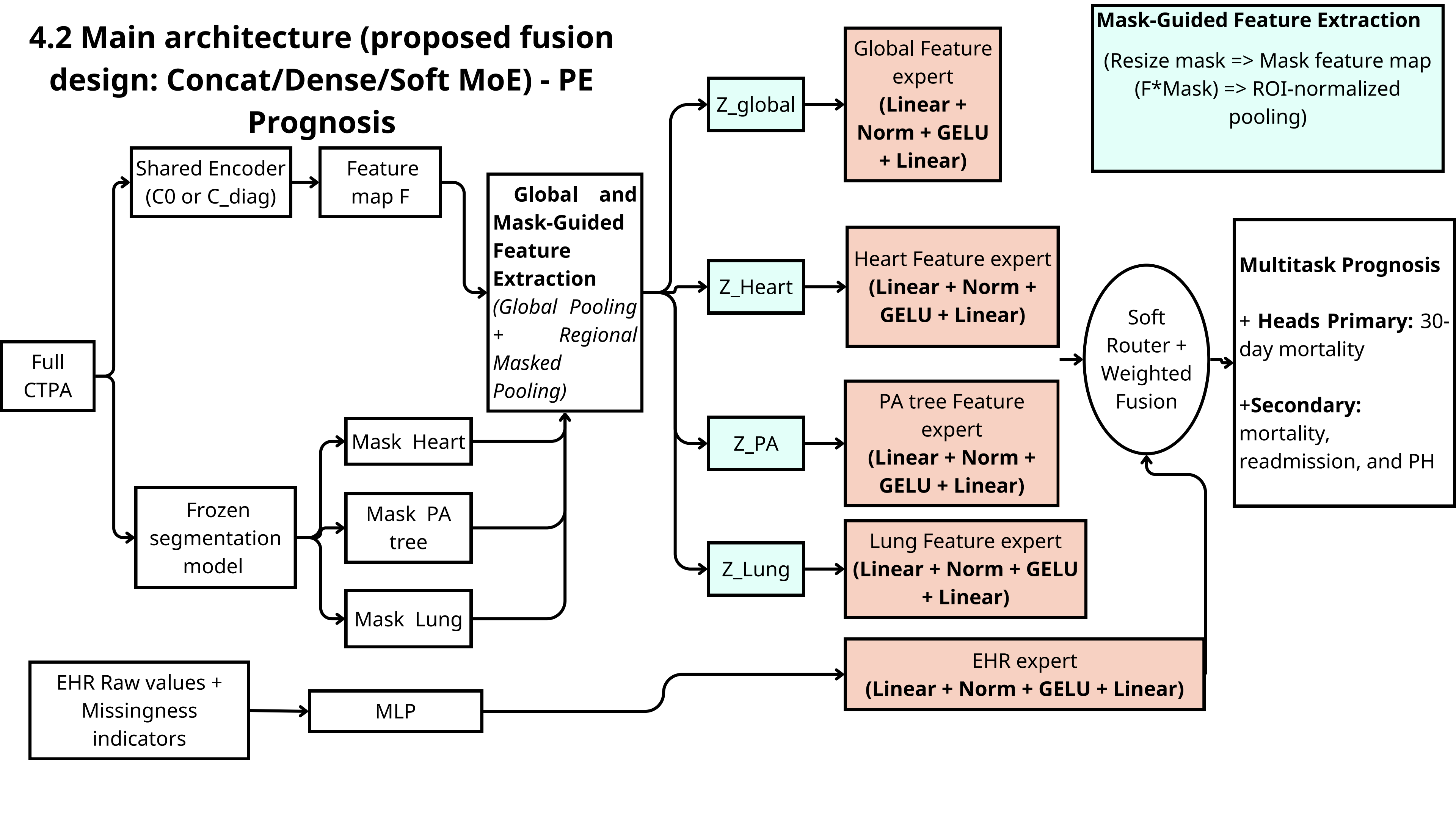


4.1 Main architecture - PE diagnosis (Conca/Dense MoE/SoftMoE)



4.2 Main architecture (proposed fusion design: Concat/Dense/Soft MoE) - PE

Prognosis



4.3 Evaluation Protocol

Diagnosis

- Patient-level.
- AUROC, AUPRC, sensitivity, specificity, F1.
- Internal testing on INSPECT.
- External testing on Turkey.
- Bootstrap 95% confidence intervals.

Prognosis

- Primary cohort: confirmed acute PE.
- AUROC, AUPRC, sensitivity and specificity.
- Bootstrap 95% confidence intervals.
- Paired comparison on the same patients.
- External testing on Turkey (if data having labels and clinical variables).

5.0 Baseline Comparisons

Diagnosis baselines

Model	Input	Main target
SwinUNETR	CTPA	PE presence
Resnet152	CTPA	PE presence
Report-only NLP baseline (MedGemma)	Report	PE presence
PENet baseline	CTPA	PE presence
Full-CT image baseline	CTPA	PE presence
Full-CT image + Finetune	CTPA	PE presence
My Proposed anatomy-aware model (without softMoE)	CTPA + masks	PE presence
My Proposed anatomy-aware model (with softMoE)	CTPA + masks	PE presence

Prognosis baselines

Model	Input
Clinical-only baseline	EHR + missingness indicators
Image-only Full-CT baseline	CTPA
Image + Clinical baseline	CTPA + EHR
Proposed anatomy-aware multimodal model (ours)	CTPA + masks + EHR + missing

Using model Proposal model MoE to evaluate impact number task.

Training variant	Labels
Single-task	Native pe_positive
Limited native multitask	pe_positive + pe_acute + pe_subsegmentalonly
Silver-label multitask	Native label (3 task) + report-derived auxiliary labels
Silver-label multitask	Native label (single task) + report-derived auxiliary labels

Expanded experiment by finetune and evaluate in RSPECT with multitask using shared encoder without using sliver label

Model	Input
SwinUNETR	CTPA
Resnet152	CTPA
PENet baseline	CTPA
Full-CT image baseline	CTPA
Full-CT image + Finetune	CTPA
My Proposed anatomy-aware model (without softMoE) no using sliver label of INSPECT	CTPA + masks
My Proposed anatomy-aware model (with softMoE) no using sliver label of INSPECT	CTPA + masks

Ablation 1: Architecture ablation in diagnosis and prognosis about number expert for MoE

Global Only	Global + Heart	Global + PAtree	Global + Lung	Full Experts with Dense or Soft MoE	Full Experts w/o Router <i>(Concatenation + Fusion MLP)</i>
-------------	----------------	-----------------	---------------	-------------------------------------	--

THANK YOU